```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, f1_score

# Load Dataset
data = load_breast_cancer()

X = data.data
y = data.target

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)

print("====================================")
print("Decision Tree using Gini Index")
print("====================================")

gini_model = DecisionTreeClassifier(
    criterion='gini',
    random_state=42
)

gini_model.fit(X_train, y_train)

gini_pred = gini_model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, gini_pred))
print("F1 Score :", f1_score(y_test, gini_pred))

print("\n====================================")
print("Decision Tree using Entropy")
print("====================================")

entropy_model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)
```

```python
entropy_model.fit(X_train, y_train)

entropy_pred = entropy_model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, entropy_pred))
print("F1 Score :", f1_score(y_test, entropy_pred))

# ------------------------------------
# Effect of Tree Depth
# ------------------------------------

depths = [2,3,4,5,6,8,10,None]

train_acc = []
test_acc = []

for d in depths:

    model = DecisionTreeClassifier(
        criterion='gini',
        max_depth=d,
        random_state=42
    )

    model.fit(X_train,y_train)

    train_acc.append(model.score(X_train,y_train))
    test_acc.append(model.score(X_test,y_test))

print("\nTree Depth Analysis")

for d,tr,te in zip(depths,train_acc,test_acc):

    print(f"Depth={d}  Train={tr:.3f}  Test={te:.3f}")

# ------------------------------------
# Cost Complexity Pruning
# ------------------------------------

path = gini_model.cost_complexity_pruning_path(X_train,y_train)

ccp_alphas = path.ccp_alphas

best_accuracy = 0
best_alpha = 0
```

```python
print("\nPruning Analysis")

for alpha in ccp_alphas:

    model = DecisionTreeClassifier(
        criterion='gini',
        ccp_alpha=alpha,
        random_state=42
    )

    model.fit(X_train,y_train)

    pred = model.predict(X_test)

    acc = accuracy_score(y_test,pred)

    if acc > best_accuracy:
        best_accuracy = acc
        best_alpha = alpha

print("Best Alpha :",best_alpha)
print("Best Accuracy :",best_accuracy)

# ------------------------------------
# Graph
# ------------------------------------

plt.figure(figsize=(8,5))
plt.plot(range(len(depths)),train_acc,marker='o',label="Train Accuracy")
plt.plot(range(len(depths)),test_acc,marker='s',label="Test Accuracy")
plt.xticks(range(len(depths)),[str(d) for d in depths])
plt.xlabel("Tree Depth")
plt.ylabel("Accuracy")
plt.title("Effect of Tree Depth")
plt.legend()
plt.show()
```

```
====================================
Decision Tree using Gini Index
====================================
Accuracy : 0.9415204678362573
F1 Score : 0.9528301886792453

====================================
Decision Tree using Entropy
====================================
Accuracy : 0.9649122807017544
F1 Score : 0.9724770642201835
```

```
====================================
Accuracy : 0.9415204678362573
F1 Score : 0.9528301886792453

====================================
Decision Tree using Entropy
====================================
Accuracy : 0.9649122807017544
F1 Score : 0.9724770642201835

Tree Depth Analysis
Depth=2  Train=0.942  Test=0.930
Depth=3  Train=0.970  Test=0.965
Depth=4  Train=0.995  Test=0.953
Depth=5  Train=0.995  Test=0.953
Depth=6  Train=0.997  Test=0.959
Depth=8  Train=1.000  Test=0.942
Depth=10  Train=1.000  Test=0.942
Depth=None  Train=1.000  Test=0.942

Pruning Analysis
Best Alpha : 0.0049148318490037
Best Accuracy : 0.9649122807017544
```